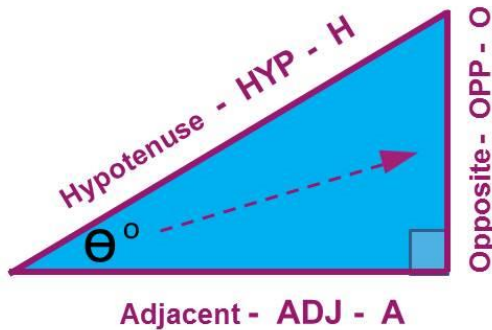


Trig Ratios – SOH CAH TOA



Name	Ratio	Expression
Sine	O / H	$\sin \Theta$
Cosine	A / H	$\cos \Theta$
Tangent	O / A	$\tan \Theta$

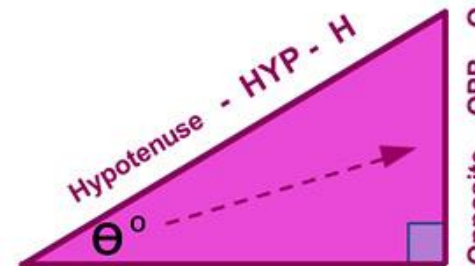
We use "SOH-CAH-TOA" to help us remember the Ratios

SOH is short for **Sine** = Opposite / Hypotenuse = O / H

CAH is short for **Cosine** = Adjacent / Hypotenuse = A / H

TOA is short for **Tangent** = Opposite / Adjacent = O / A

Sine Ratio – All Formulas



$$\sin \Theta = \frac{\text{OPP}}{\text{HYP}}$$

$$\text{OPP} = \text{HYP} \times \sin \Theta$$

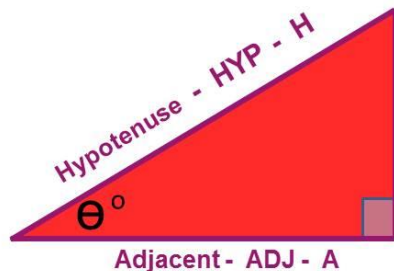
$$\text{HYP} = \frac{\text{OPP}}{\sin \Theta}$$

These are the four formulas for working with Sine Triangles.

We also use the special "Sin" and "Sin⁻¹" calculator buttons when solving Sine Triangles.

$$\Theta = \sin^{-1} \frac{\text{OPP}}{\text{HYP}}$$

Cosine Ratio – All Formulas



$$\cos \Theta = \frac{\text{ADJ}}{\text{HYP}}$$

$$\text{ADJ} = \text{HYP} \times \cos \Theta$$

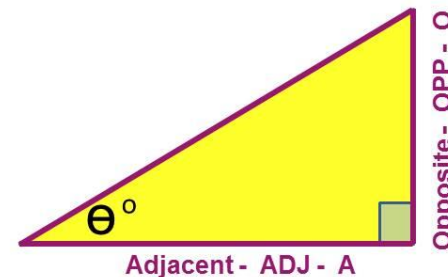
$$\text{HYP} = \frac{\text{ADJ}}{\cos \Theta}$$

These are the four formulas for working with Cosine Triangles.

We also use the special "Cos" and "Cos⁻¹" calculator buttons when solving Cosine Triangles.

$$\Theta = \cos^{-1} \frac{\text{ADJ}}{\text{HYP}}$$

Tangent Ratio – All Formulas



$$\tan \Theta = \frac{\text{OPP}}{\text{ADJ}}$$

$$\text{OPP} = \text{ADJ} \times \tan \Theta$$

$$\text{ADJ} = \frac{\text{OPP}}{\tan \Theta}$$

These are the four formulas for working with Tangent Triangles.

We also use the special "Tan" and "Tan⁻¹" calculator buttons when solving Tangent Triangles.

$$\Theta = \tan^{-1} \frac{\text{OPP}}{\text{ADJ}}$$